CS512 Machine Learning, Fall 2025: Homework 1

Due: Nov 15 2026, 22:00

Instructions

- Use Python for all programming tasks.
- Submit your report, code, and data files together as a single zip file on SUCourse:
  - **Report:** A PDF containing all of your answers, along with the relevant code outputs (e.g., plots). Name it `CS512-HW1-YourName.pdf`.
  - **Code:** A Jupyter notebook. Run all cells before submitting so that every output is visible. Name it `CS512-HW1-YourName.ipynb`.
  - **Data:** Any data files needed to run your notebook.

  Replace `YourName` in both filenames with your first and last names (e.g., `CS512-HW1-AyseYilmaz.pdf`).
- Late submissions are handled according to the late submission policy in the syllabus, based on your submission timestamp and remaining late days.
- Submissions that do not follow these instructions (e.g., wrong file names, missing outputs, or missing files) will receive point deductions.

1 AI-assisted Image Annotation [10 pts]

A company uses an AI-assisted annotation tool to assign a single label to each image in a dataset. Each image is then reviewed by exactly one annotator from one of three teams: A, B, and C. After review, each image has one final label, which is either correct or incorrect.

The teams differ in size and in their error rates after review:

- Team A employs 50 annotators. After review by this team, 4% of the images have an incorrect final label.
- Team B employs 30 annotators. After review by this team, 10% of the images have an incorrect final label.
- Team C employs 20 annotators. After review by this team, 25% of the images have an incorrect final label.

Each image is equally likely to be assigned to any annotator across the three teams. Thus, the probability that an image is reviewed by a given team is proportional to the number of annotators in that team. After all images have been reviewed, a quality-control specialist selects one image uniformly at random from the dataset.

a) [5 pts] Given that the selected image has an incorrect final label, what is the probability that it was reviewed by team A?

b) [5 pts] Given that the selected image has a *correct final label*, what is the probability that it was reviewed by team A?

2 Spelling Correction [10 pts]

a) [8 pts] A smartphone company implements a simple spelling corrector based on Bayes' rule. Suppose a user types `form`. The intended word could be `from`, `foam`, or `form` itself.

The relative frequencies of these words among all words in a large text corpus are provided in Table 1. Assume that the intended word is one of these three candidates. Determine the prior probabilities by normalizing the corpus frequencies over these three candidates.

Word	Relative frequency in the corpus
`from`	4.00×10^{-3}
`form`	1.00×10^{-4}
`foam`	1.00×10^{-5}

Table 1: Relative frequencies of the three candidate words in a large corpus.

Typing-error data provide the following information: i) when the intended word is `from`, the user types `form` 2% of the time; ii) when the intended word is `foam`, the user types `form` 0.4% of the time; and iii) when the intended word is `form`, the user types it correctly 98% of the time.

Given that the user typed `form`, calculate the posterior probability of each of the three possible intended words. Which word should the corrector select if it chooses the candidate with the highest posterior probability? Show your work.

b) [2 pts] A simpler spelling corrector always selects the most frequent of the three candidate words in the table, including the typed word itself. Would it select the same word as the Bayesian corrector in part (a)? Explain your answer and the role of typing-error probabilities in two sentences.

3 Thumbtacks Revisited [16 pts]

Maximum Likelihood Estimation

Let's go back to our thumbtack example in class. θ is the probability of a thumbtack falling on heads.

a) [4 pts] Consider two datasets: one contains 6 heads and 4 tails, and the other contains 60 heads and 40 tails.

Calculate the MLE of θ for each dataset. Plot both likelihood functions on the same axes, dividing each likelihood by its maximum value so that both curves have a maximum height of one.

Compare the locations of the maxima and the concentration of the curves around them. What does this comparison tell you that the MLE values alone do not?

Maximum a Posteriori Estimation

In Bayesian estimation, we express our prior knowledge about θ using a prior probability density $p(\theta)$. Conditional on θ, assume that the observations $X_1, \ldots, X_N$ are independent Bernoulli random variables, where $X_i = 1$ represents heads and $X_i = 0$ represents tails.

After observing the data $D = (X_1, \ldots, X_N)$, we update our knowledge using Bayes' rule:

$$p(\theta \mid D) = \frac{P(D \mid \theta)\, p(\theta)}{\displaystyle\int_0^1 P(D \mid u)\, p(u)\, du}.$$

The maximum a posteriori (MAP) estimate is the value of θ that maximizes the posterior density:

$$\hat{\theta}_{\text{MAP}} = \arg\max_{\theta \in [0,1]} p(\theta \mid D).$$

Because the denominator in Bayes' rule does not depend on θ, we can equivalently write

$$\hat{\theta}_{\text{MAP}} = \arg\max_{\theta \in [0,1]} L(\theta)\, p(\theta),$$

where $L(\theta) = P(D \mid \theta)$ is the likelihood. Thus, MAP estimation combines the likelihood with the prior density.

The Beta distribution. A random variable θ with a Beta(a, b) distribution has density

$$p(\theta \mid a, b) = \frac{1}{B(a,b)} \theta^{a-1}(1-\theta)^{b-1}, \qquad 0 < \theta < 1,$$

where $a, b > 0$, and

$$B(a,b) = \int_0^1 t^{a-1}(1-t)^{b-1}\, dt = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}.$$

The beta function $B(a, b)$ is the normalization constant that ensures the density integrates to one.

b)) [6 pts] Suppose that N independent thumbtack tosses produce α_H heads and α_T tails, where $N = \alpha_H + \alpha_T$. Assume the prior

$$\theta \sim \text{Beta}(\beta_H, \beta_T), \qquad \beta_H, \beta_T > 0.$$

Derive the normalized posterior density $p(\theta \mid D)$. Identify the distribution family and its parameters. Show your derivation, including how the likelihood and prior combine.

c) [6 pts] Assume a Beta$(3, 3)$ prior. As in part (a), suppose that $N = 10$ tosses produce 6 heads and 4 tails. Calculate the MAP estimate analytically and compare it with the MLE from part (a). Briefly explain how the prior influences the estimate.

4 Building a Spam Classifier with Naive Bayes [60 pts]

Your job is to build a spam classifier that can predict whether an email is spam or not. The questions summarize the model; therefore, please read all the questions before starting coding.

IMPORTANT:

- **Permitted libraries:** You may use NumPy and pandas for loading data and performing numerical operations, and Matplotlib for plotting. You must implement parameter estimation, log-score computation, and prediction yourself. Do not use a library implementation of Naive Bayes.
- **Training and evaluation:** Estimate all model parameters using only `train-features.txt` and `train-labels.txt`. Evaluate your classifiers using `val-features.txt` and `val-labels.txt`. Do not use the validation data to estimate model parameters.
- **Handling zero probabilities:** When computing log scores, a word that does not appear in an email contributes zero, even if its estimated probability is zero. Thus, use the convention $0 \log 0 = 0$. If a word appears at least once but has zero estimated probability under a class, that class receives a score of $-\infty$. Handle these cases explicitly to avoid producing `NaN` values.
- **Handling ties:** If the two classes have equal log scores, predict non-spam (label 0). This rule also applies when both scores are $-\infty$.
- **Reporting results:** Include all requested numerical results, plots, tables, and explanations in your PDF report. The report must be understandable on its own, without requiring the reader to open your notebook.

Dataset

Your dataset is a preprocessed and modified subset of the Ling-Spam Dataset [1]. It is based on 960 real email messages from a linguistics mailing list. Emails have been preprocessed in the following ways:

- **Stop word removal:** Words like "and","the", and "of", are very common in all English sentences and are therefore not very predictive in deciding spam/nonspam status. These words have been removed from the emails.
- **Lemmatization:** Words that have the same meaning but different endings have been adjusted so that they all have the same form. For example, "include", "includes," and "included," would all be represented as "include." All words in the email body have also been converted to lower case.
- **Removal of non-words:** Numbers and punctuation have both been removed. All white spaces (tabs, newlines, spaces) have all been trimmed to a single space character

The data has already been split into two subsets: a 700-email subset for training and a 260-email subset for testing (consider this as your validation set and imagine there is another test set which is not given to you). The features have been generated for you. You will use the following files:

- `train-features.txt`
- `train-labels.txt`
- `val-features.txt`
- `val-labels.txt`

The files ending with `features.txt` contain the features, and the files ending with `labels.txt` contain the ground-truth labels.

In the feature files, each row contains the feature vector for an email. The j-th entry in row i is the number of occurrences of the j-th vocabulary word in the i-th email. The vocabulary contains 2500 words. The label files contain the corresponding ground-truth labels, with emails appearing in the same order as in the feature files. Thus, the i-th row in each pair of files corresponds to the same email. Label 1 stands for spam, and label 0 stands for non-spam.

Bag-of-Words Representation and Multinomial Naive Bayes Model

Recall that the bag-of-words representation ignores word order. Multinomial Naive Bayes additionally assumes that word tokens are conditionally independent given the email class, and that their class-conditional distribution is the same at every position. We treat email length as given and do not model it as class-dependent.

For an email D_i containing n_i word tokens from the vocabulary, the probability of its observed word sequence, given class y_k, is

$$\mathbf{P}\left(D_i \,|\, Y = y_k\right) = \mathbf{P}\left(X_1 = x_1, \ldots, X_{n_i} = x_{n_i} \,|\, Y = y_k\right) = \prod_{j=1}^{n_i} \mathbf{P}\left(X_j = x_j \,|\, Y = y_k\right). \tag{4.1}$$

In Eq. (4.1), X_j represents the word token at the j-th position in email D_i, and x_j is its observed value. The number of positions is n_i. For example, the first email D_1 might contain 200 word tokens ($n_1 = 200$), belong to the spam class ($y_k = 1$), and contain the word "office" at its 15th position ($x_{15} =$ "office").

In the above formulation, the sequence length depends on the number of word tokens in the email. Thus, different emails may have different sequence lengths. This differs from our feature files, where

the j-th entry in row i is the number of occurrences of the j-th vocabulary word in email i. We can group repeated word occurrences to obtain

$$\mathbf{P}\left(D_i \,|\, Y = y_k\right) = \prod_{j=1}^{V} \mathbf{P}\left(X = w_j \,|\, Y = y_k\right)^{t_{w_j,i}}, \tag{4.2}$$

where V is the vocabulary size, X denotes a generic word token, and $t_{w_j,i}$ is the number of occurrences of vocabulary word w_j in email D_i.

For example, suppose $V = 2500$, the first email D_1 is spam ($y_k = 1$), and the 80th vocabulary word, w_{80}, is "click". Then $t_{w_{80},1} = 2$ means that "click" appears twice in D_1. Consider why Eq. (4.1) and Eq. (4.2) are equivalent.

Here, D_i denotes the observed word sequence, so no multinomial coefficient is included. The probability of the count vector itself would include the coefficient $n_i!/\prod_{j=1}^{V} t_{w_j,i}!$. This coefficient is independent of the class and cancels when computing posterior class probabilities.

In classification, we are interested in the probability distribution over the email classes given an email D_i. Bayes' rule gives

$$\mathbf{P}\left(Y = y_k \,|\, D_i\right) = \frac{\mathbf{P}\left(Y = y_k\right) \prod_{j=1}^{V} \mathbf{P}\left(X = w_j \,|\, Y = y_k\right)^{t_{w_j,i}}}{\sum_{\ell=0}^{1} \mathbf{P}\left(Y = y_\ell\right) \prod_{j=1}^{V} \mathbf{P}\left(X = w_j \,|\, Y = y_\ell\right)^{t_{w_j,i}}}, \tag{4.3}$$

where $y_0 = 0$ denotes non-spam and $y_1 = 1$ denotes spam.

For classification, we can ignore the denominator because it is the same for both classes:

$$\mathbf{P}\left(Y = y_k \,|\, D_i\right) \propto \mathbf{P}\left(Y = y_k\right) \prod_{j=1}^{V} \mathbf{P}\left(X = w_j \,|\, Y = y_k\right)^{t_{w_j,i}}. \tag{4.4}$$

Therefore,

$$\hat{y}_i = \arg\max_{y_k \in \{0,1\}} \mathbf{P}\left(Y = y_k\right) \prod_{j=1}^{V} \mathbf{P}\left(X = w_j \,|\, Y = y_k\right)^{t_{w_j,i}}. \tag{4.5}$$

Multiplying many small probabilities can cause numerical underflow. We therefore compute log scores instead. Because the logarithm is strictly increasing for positive arguments, taking logarithms does not change which class maximizes a positive score:

$$\hat{y}_i = \arg\max_{y_k \in \{0,1\}} \left(\log \mathbf{P}\left(Y = y_k\right) + \sum_{j=1}^{V} t_{w_j,i} \log \mathbf{P}\left(X = w_j \,|\, Y = y_k\right) \right). \tag{4.6}$$

Here, $\hat{y}_i$ is the predicted label for email i.

a) [2 pts] A dataset has a balanced class distribution if its classes contain approximately equal numbers of examples; otherwise, its class distribution is imbalanced. What percentage of the emails in `train-labels.txt` are spam? Is the training set balanced or skewed toward one of the classes?

The MLEs of the model parameters are

$$\hat{\theta}_{j|y=0} = \frac{T_{j,0}}{\sum_{\ell=1}^{V} T_{\ell,0}}, \qquad \hat{\theta}_{j|y=1} = \frac{T_{j,1}}{\sum_{\ell=1}^{V} T_{\ell,1}},$$

$$\hat{\pi}_{y=1} = \frac{N_1}{N}, \qquad \hat{\pi}_{y=0} = 1 - \hat{\pi}_{y=1}.$$

- $T_{j,0}$ is the total number of occurrences of word w_j in non-spam training emails, including repeated occurrences within an email.
- $T_{j,1}$ is the total number of occurrences of word w_j in spam training emails, including repeated occurrences within an email.
- N_1 is the number of spam training emails.
- N is the total number of training emails.
- $\hat{\pi}_{y=1}$ estimates the probability that an email is spam.
- $\hat{\theta}_{j|y=0}$ estimates the probability that a word token in a non-spam email is vocabulary word w_j, $\mathbf{P}(X = w_j \,|\, Y = 0)$.
- $\hat{\theta}_{j|y=1}$ estimates the probability that a word token in a spam email is vocabulary word w_j, $\mathbf{P}(X = w_j \,|\, Y = 1)$.

b) [5 pts] How many free (independent) parameters do we need to estimate for this model? Account for the normalization constraints.

c) [40 pts] Train a Naive Bayes classifier using all the training data (`train-features.txt` and `train-labels.txt`). Evaluate your classifier on the validation data (`val-features.txt` and `val-labels.txt`). Report the validation accuracy and the number of incorrect predictions. Use the MLEs above to estimate the parameters.

Important: Use the conventions $0 \log 0 = 0$ and $a \log 0 = -\infty$ for $a > 0$. Handle these cases explicitly to avoid producing `NaN` values. In case of equal class scores, predict non-spam. This rule also applies when both scores are $-\infty$.

d) [8 pts] Extend your classifier to compute MAP estimates of the θ parameters using a symmetric Dirichlet prior Dirichlet$(\alpha+1, \ldots, \alpha+1)$, where all V prior parameters are equal to $\alpha+1$. This gives additive smoothing by adding α to each word count within each class:

$$\hat{\theta}^{\text{MAP}}_{j|y=0} = \frac{T_{j,0} + \alpha}{\sum_{\ell=1}^{V} T_{\ell,0} + \alpha V},$$

$$\hat{\theta}^{\text{MAP}}_{j|y=1} = \frac{T_{j,1} + \alpha}{\sum_{\ell=1}^{V} T_{\ell,1} + \alpha V}.$$

Continue to estimate the class priors using MLE.

For this question, set $\alpha = 1$, corresponding to a Dirichlet$(2, \ldots, 2)$ prior. Train using all the training data, evaluate on the validation data, and report the validation accuracy. Compare your result with part (c) and comment on the difference.

e) [5 pts] Using the smoothed estimates from part (d), rank the features by the absolute value of their log-ratio:

$$r_j = \log \frac{\hat{\theta}^{\text{MAP}}_{j|y=1}}{\hat{\theta}^{\text{MAP}}_{j|y=0}}.$$

Report the top five features overall in a table, including their vocabulary indices (using indices $1, \ldots, 2500$), signed log-ratios, and the classes they favor. A positive log-ratio favors spam, a negative log-ratio favors non-spam, and zero indicates no preference.

References

1. Liang Spam dataset. http://csmining.org/index.php/ling-spam-datasets.html
2. "On Discriminative vs. Generative Classifiers: A comparison of logistic regression and Naive Bayes" by Andrew Ng and Michael I. Jordan.
3. Manning, C. D., Raghavan, P., and Schutze, H. (2008). Introduction to information retrieval. New York: Cambridge University Press.
http://nlp.stanford.edu/IR-book/html/htmledition/mutual-information-1.html
4. CMU Lecture Notes.
http://www.cs.cmu.edu/~epxing/Class/10701-10s/Lecture/lecture5.pdf